

RADIO DIRECTION FINDER

NEED

The study of Radio Direction Finding (RDF) is important in the military because it helps locate and track enemy radio transmissions. This capability supports intelligence gathering, enhances situational awareness, improves force protection, and assists commanders in making informed operational decisions.

TIME



ROADMAP

- **Diagnose** the following:
  - RDF role and radio frequency sources.
  - RDF concepts.
  - Methods used in RDF to determine direction of arrival (DOA)
  - Components of an RDF system.
  - Setting up and calibrating an RDF system.
  - RDF system parameters.
  - RDF measurements.
  - RDF readings and bearing.
  - Methods to mitigate errors and enhance accuracy in RDF measurements.
  - RDF techniques and radio frequency emitters.
  - RDF operations with other intelligence.
  - RDF operations in achieving mission objectives.
- **Experiment:** check on learning
- **Reflect:** takeaways on the lesson
- **Evaluate:** Written Examination (Single Response/Tru or False)

REFERENCE

- PAM 6-00 (Command and Control Communications and Cyber Systems)
- PAM 6-0102 – Signal Operations
- PAM 3-00 (Army Operations)
- PAM 6-0102-4 Cyberspace Operations
- FM 3-12- Cyberspace and Electronic Warfare Operations

SAFETY PRECAUTION



The nearest person on the fire extinguisher will extinguish and control the fire. The facilitator will be the last person to leave the classroom. Consolidation point will be Infront of Yan Hall.

SAFETY PRECAUTION



In case of earthquake: **“DROP, COVER AND HOLD ON.”** and the emergency evacuation plan before the start of the training and ensure that the learner understands.



SAFETY PRECAUTION



Learner to be observant of broken electrical wires.

“SAFETY in mind, leave accidents behind.”



★ ★ — ARMY 2040: WORLD-CLASS. MULTI-MISSION READY. CROSS-DOMAIN CAPABLE. — ★ ★

HEAT INDEX SAFETY



Due to the high heat index being experienced nationwide, the Office of the Chief Surgeon reiterates the importance of observing heat safety measures during training and outdoor activities.
All personnel are enjoined to strictly follow the following guidelines.

	SAFETY MEASURE	GUIDELINES
1	STAY HYDRATED	• Drink plenty of water even if not thirsty. • Avoid sugary, caffeinated, and alcoholic drinks. • Hydrate before, during, and after activities to maintain optimal performance and alertness.
2	SEEK PROPER SHELTER	• Take frequent breaks under shaded or covered areas. • Use hats, caps, or umbrellas to protect from direct sun exposure.
3	PACE YOURSELF FOR THE DAY	• Adjust your pace with rest cycles in mind. • Slow down and avoid overexertion. • Build up your activity gradually while closely monitoring your body's response.
4	WEAR COOL, LIGHTWEIGHT CLOTHING	• Lightweight, breathable, moisture-wicking clothing helps regulate body temperature. • Light-colored clothing reflects sunlight and helps keep cool.
5	MONITOR FELLOWS (BUDDY SYSTEM)	• Observe one another for signs of heat exhaustion or heat stroke such as dizziness, confusion, excessive sweating, and weakness. • Report early. Report and assist anyone showing symptoms.
6	IMMEDIATE ACTION IF HEAT ILLNESS OCCURS	• Stop activity immediately. • Move to a cool, shaded area. • Provide water or oral rehydration (if not nauseated). • Provide first aid and seek medical attention.
7	ASSOCIATE COMMANDER SUPERVISION	• Ensure leaders are heat-index aware, maintain appropriate work-rest cycles, personnel with chronic medical conditions, and those under medication or vitamin supplements. • Leaders shall disseminate and enforce health and heat safety protocols.


THINK
OUR AIM . . .
NO ACCIDENTS

CLASSROOM RULES













RADIO DIRECTION FINDER

RDF technology is critical for the Philippine Army to enhance its operational capabilities, ensure national security, support disaster response efforts, and prepare for various military and humanitarian missions effectively.

Principles of Radio Direction Finder

RDF role and radio frequency sources

Role of RDFs:

Locating RF Sources:

- **Military Applications:** RDFs are extensively used in military operations to locate enemy communication centers, radar installations, and other electronic emissions
- **Civilian Applications:** In civilian contexts, RDFs can be used for spectrum monitoring, identifying sources of interference, and locating emergency distress signals (e.g., from aircraft, ships, or distress beacons)

Principles of Radio Direction Finder

RDF role and Radio Frequency Sources

Role of RDFs:

Navigation and Positioning:

- Historically, RDFs have been used for navigation, especially in marine and aviation applications where they help determine the direction of radio beacons or signals, aiding in establishing position without relying on visual landmarks or GPS

Principles of Radio Direction Finder

RDF role and Radio Frequency Sources

Role of RDFs:

Search and Rescue:

- RDFs are invaluable in search and rescue operations to pinpoint the location of distress signals, guiding responders to the precise location of survivors or downed aircraft in remote or inaccessible areas

Principles of Radio Direction Finder

RDF role and Radio Frequency Sources

Role of RDFs:

Monitoring and Surveillance:

- Governments use RDF technology for monitoring RF emissions in border security, law enforcement, and intelligence operations. RDFs can detect unauthorized transmissions, monitor communication networks, and enforce radio spectrum regulations

Principles of Radio Direction Finder

RDF role and Radio Frequency Sources

Radio Frequency Sources:

Military Communications:

- RDFs are used to intercept and locate military communications, including tactical radios, satellite communications, and radar signals. This information is critical for military intelligence and reconnaissance operations

Principles of Radio Direction Finder

RDF role and Radio Frequency Sources

Radio Frequency Sources:

Radar Systems:

- Radar installations emit distinctive RF signals that can be detected and located using RDFs. This capability allows military forces to identify enemy radar sites, assess their capabilities, and potentially neutralize them

Principles of Radio Direction Finder

RDF role and Radio Frequency Sources

Radio Frequency Sources:

Civilian Communications:

- RDFs can detect and locate civilian communication sources such as radio and television broadcasting stations, cellular networks, and other licensed or unauthorized radio transmissions

Principles of Radio Direction Finder

RDF role and Radio Frequency Sources

Radio Frequency Sources:

Emergency Beacons:

- RDFs are used to locate distress beacons (Emergency Position-Indicating Radio Beacons, or EPIRBs) used by aircraft, ships, and individuals in distress situations. This capability is crucial for coordinating search and rescue efforts

Principles of Radio Direction Finder

RDF concepts

Intelligence Gathering:

- RDF technology is used by the Philippine Army to intercept and locate enemy communication signals. This capability helps in gathering intelligence about the movements, positions, and intentions of adversaries

Principles of Radio Direction Finder

RDF concepts

Communication Monitoring:

- RDFs are employed to monitor and analyze radio transmissions within the operational area. This includes detecting unauthorized or suspicious communications, as well as ensuring the security of military communications

Principles of Radio Direction Finder

RDF concepts

Border Surveillance:

- RDF technology assists in monitoring and securing the country's borders. By detecting and locating radio signals originating from unauthorized or suspicious sources near borders, the Philippine Army can enhance border surveillance and control

Principles of Radio Direction Finder

RDF concepts

Search and Rescue:

- RDFs are essential in search and rescue operations conducted by the Philippine Army. They can locate distress signals from downed aircraft, stranded personnel, or individuals in remote or difficult-to-access areas, facilitating timely rescue missions

Principles of Radio Direction Finder

RDF concepts

Training and Capability Development:

- Training personnel in RDF operations is crucial for maintaining proficiency and readiness within the Philippine Army. This includes understanding RDF principles, techniques for signal triangulation, and effective utilization of RDF equipment in diverse operational scenarios

Principles of Radio Direction Finder

RDF concepts

Integration with Command and Control Systems:

- RDF technology is integrated into broader command and control systems used by the Philippine Army. This integration enhances the Army's ability to coordinate operations, manage resources, and respond rapidly to changing situations based on real-time intelligence gathered from RDF operations

Principles of Radio Direction Finder

Methods used in RDF to determine direction of arrival (DOA)

Phase Comparison (Differential RDF):

- **Concept:** Differential RDF compares the phase of the signal received at two or more separate antennas

- **Operation:** The phase difference between signals received by different antennas is measured. This phase difference is used to determine the direction of the RF source

Principles of Radio Direction Finder

Methods used in RDF to determine direction of arrival (DOA)

Time Difference of Arrival (TDOA):

- **Concept:** TDOA methods involve using the difference in arrival times of the RF signal at multiple spatially separated antennas

- **Operation:** By precisely measuring the time it takes for the RF signal to arrive at each antenna, the RDF system calculates the differences in arrival times

Principles of Radio Direction Finder

Methods used in RDF to determine direction of arrival (DOA)

Interferometric Techniques:

- **Concept:** Interferometric RDF methods use the phase differences of signals received by multiple antennas to determine the DOA

- **Operation:** By comparing the phase differences between signals received at different antenna pairs or elements, interferometric RDF systems can accurately determine the azimuth of the RF source

RDF Measurements and Analysis RDF measurements

Direction of Arrival (DOA):

Definition: DOA is the azimuthal angle indicating the direction from which the RF signal is arriving relative to the RDF system.

RDF operations to practical scenarios

RDF operations in achieving mission objectives

Military Applications:

Intelligence and Surveillance:

- **Objective:** Gather intelligence by locating enemy communication signals, radar installations, or electronic warfare (EW) activities

- **RDF Contribution:** RDF enables military forces to identify and monitor adversary movements, positions, and intentions based on intercepted RF emissions. This intelligence aids in planning and executing tactical operations with enhanced situational awareness.

SUMMARY

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LEARNING OUTCOME

By this time, the students will be able to apply the concept of RDF during signal operations IAW PAM 6-0102-03. (C3)

FEEDBACK





THANK YOU

